

Chapter 9, Problem 5P

Problem

Given $v_1 = 45 \sin(\omega t + 30^\circ)$ V and $v_2 = 50 \cos(\omega t - 30^\circ)$ V, determine the phase angle between the two sinusoids and which one lags the other.

Step-by-step solution

Step 1 of 3

Consider the following data:

The first sinusoidal voltage, $v_1 = 45 \sin(\omega t + 30^\circ)$ V (1)

The second sinusoidal voltage, $v_2 = 50 \cos(\omega t - 30^\circ)$ V (2)

Step 2 of 3

From equation 1,

$$\begin{aligned} v_1 &= 45 \sin(\omega t + 30^\circ) \text{ V} \\ &= 45 \cos(\omega t + 30^\circ - 90^\circ) \text{ V} \\ &= 45 \cos(\omega t - 60^\circ) \text{ V} \dots\dots (3) \end{aligned}$$

Step 3 of 3

Compare equation 2 and equation 3, it is clear that phase angle difference is:

$$\begin{aligned} \phi &= 60^\circ - 30^\circ \\ &= 30^\circ \end{aligned}$$

Therefore, the phase angle between the two sinusoids is 30° and v_1 lags v_2 .